Module 03 — Object-Oriented Programming (OOP)

Phase: Advanced Python
Previous module: Modules and Packages
Next module: Exception Handling (after Part 4)

Module Outcome:

- Part 1 — classes and encapsulation
- Part 2 — inheritance and composition
- Part 3 — polymorphism and BankAccount CLI
- Part 4 — abstract classes, interfaces, Campus Wallet

This module is split for slow-pace teaching (Functions-style depth: Motivation → Definition/Purpose → concepts with **Try these** → annotated examples → practice → project). Use the parts in order:

Part	File	Focus
1	03_OOP_Part1_Classes_Encapsulation.md	Dict pain → class/`self`/`__init__`, variables, methods, encapsulation, BankAccount lab
2	03_OOP_Part2_Inheritance.md	Inheritance basics → types (diagrams) → override → then HAS-A / composition
3	03_OOP_Part3_Polymorphism_Project.md	Polymorphism, `__str__`, CLI mini-project push, interview Q&A
4	03_OOP_Part4_Abstract_Classes_Interfaces.md	Abstract method, abstract class (`ABC`), interface (ABC-as-contract / Protocol), Campus Wallet

Lab kits: 17_bank_account_oop (start in Part 1; push in Part 3) · 32_campus_wallet_abc (Part 4)

Assignment: Part 3 has the BankAccount CLI **Mini-Project (Assignment)** — Parts 1–2 point there. Part 4 has its own Campus Wallet assignment.

Course order note: Exception Handling is **after** OOP (including Part 4). All OOP demos/labs validate with **`print` + `return`** (no `raise` / `try` / `except` required in OOP).

Optional enrichment: Professional Python Deep Dive (composition patterns, `__repr__` — ABC is taught in Part 4)